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Mechatronics Design and Build 2: R2R Spotter Plotter



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Chapter 1

Introduction

Roll-to-roll (R2R) manufacturing systems enable scalable and continuous processing of flexible substrates and are increasingly used in applications involving controlled liquid deposition. In such processes, the size, shape, and placement of individual droplets are critical quality indicators, as deviations may signal issues such as unstable flow conditions, nozzle malfunction, or process drift.

This project builds on previous work conducted on the R2R Spotter Plotter platform, where a piezoelectric jet-valve-based non-contact dispensing module was developed for precise droplet deposition [1]. While the existing system demonstrated the feasibility of high-resolution spotting, it also revealed practical limitations related to system complexity, sensitivity to operating conditions, and fluid handling robustness.

Motivated by these findings, the present work explores the development of an alternative dispensing concept based on pneumatic actuation, integrated into the same R2R framework. The goal is not to replace the existing piezo-driven dispenser, but to extend the platform by investigating a simpler and potentially more robust dispensing approach that can operate under comparable conditions.

In parallel, a computer vision-based monitoring system is developed to detect and analyze deposited droplets in real time. Due to spatial constraints, the camera is mounted at an oblique angle relative to the substrate, introducing perspective distortion and non-circular droplet appearances. To compensate for this, a reference grid with known geometry is used to map image-space measurements to physical dimensions. A deep learning-based segmentation model is employed for droplet localization, as the transparent and refractive nature of liquid droplets makes precise boundary definition challenging.

Overall, this project contributes to the continued development of the R2R Spotter Plotter by extending its dispensing capabilities and providing a robust vision-based framework for droplet detection and quality assessment.

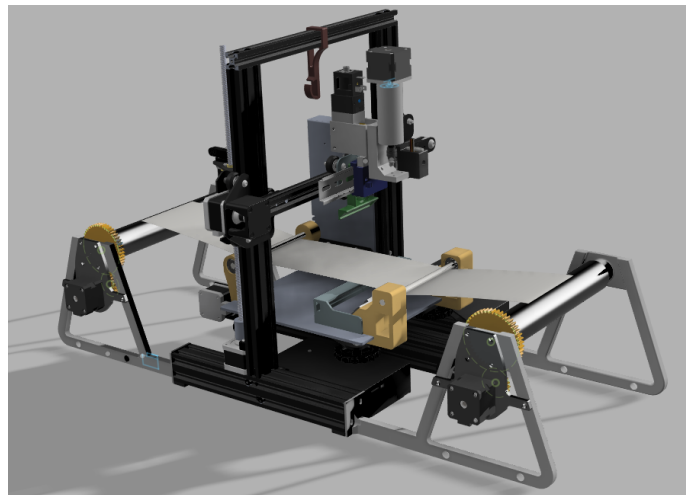


Figure 1.1: R2R full assembly

Chapter 2

Problem statement

2.1 Problem statement

The previous R2R Spotter Plotter demonstrated a proof of concept for scalable, non-contact chemical dispensing but faced challenges related to droplet size variation, pressure instability, and limited real-time control. The piezo-based actuation was effective, yet costly and difficult to tune for different fluids. To make the system more accessible and reliable, a new pneumatic-based drop generation system with solenoid control will be designed. In addition, enhanced vision-based analysis and feedback mechanisms will be implemented to make the distribution process measurable, adjustable, and repeatable.

2.2 Objectives

- Number of modular dispensing modules ≥ 2
- Fill an A4 page within 2.5 hours with 600 μm grid spacing.
- Generate droplets with a diameter range of 300 – 1500 μm with less than $\pm 10\%$ variation.
- Provide real-time droplet monitoring using a camera-vision system, to ensure correct droplet size.
- Allow adjustment of key parameters (such as timing, spacing) via a graphical user interface on PC.
- Cost per unit should not exceed 3.500 DKK (excluding module).
- Cost per dispensing module should not exceed 2.000 DKK.

2.3 Delimitations

- The project focuses only on pneumatic actuation; piezoelectric and thermal ink-jet methods are excluded.
- The system does not aim to achieve commercially available industrial-grade efficiency or accuracy.
- Fluid property optimization (e.g., for different viscosities) will not be a primary focus; a standard polymer solution will be used for testing.
- Advanced control techniques (e.g., AI-based feedback) may be discussed but not fully implemented due to time constraints.

Chapter 3

Methodology

3.1 Project Workflow Overview

The conceptual design of the system was developed using Autodesk Inventor 2026, which was used to model and evaluate the mechanical components prior to fabrication. Rapid prototyping was primarily carried out through additive manufacturing, using PLA and PETG materials to enable fast iteration and design refinement. The project was structured into several distinct but interconnected development stages. These included the mechanical design of the dispensing system, the development of a metal-based dispenser bottom unit, the roll-to-roll transport mechanism, and modular mounting solutions to facilitate system integration. In parallel, the electronic control architecture and the associated software infrastructure were developed, including motion control, user interfaces, and AI-based droplet detection. The initial schedule is represented in the Gantt Chart in Figure 3.1.

3.2 Integration strategy

Each subsystem was designed, fabricated, and tested independently to verify functionality and performance before being integrated into the complete system. This modular and iterative workflow reduced integration risk and enabled systematic identification and resolution of design issues throughout the development process.

3.3 Subsystem Testing and Validation

Each subsystem was validated using functional testing before integration. Mechanical components were evaluated for structural stability, alignment accuracy, and repeatability of motion. The roll-to-roll mechanism was tested for consistent substrate transport and sufficient tensioning under operating conditions. Electronic and control subsystems were verified through motion commands, sensor feedback, and communication reliability. For the software and AI components, validation focused on correct system operation, real-time performance, and the ability to detect and measure droplets consistently under controlled conditions. Only after meeting predefined functional criteria were subsystems integrated into the full system.

3.4 Iterative Development Approach

The development followed an iterative design approach, where design decisions were continuously refined based on testing outcomes. Feedback from subsystem testing informed subsequent design modifications, allowing mechanical, electronic, and software components to co-evolve throughout the project. This approach enabled rapid identification of design limitations and supported incremental system improvement.

3.5 Project Constraints and Assumptions

The methodology was influenced by practical constraints, including limited project duration, hardware availability, and partial functionality of the dispensing system during later development stages. Additionally, AI model training relied on data collected from a prototype setup rather than the final system.

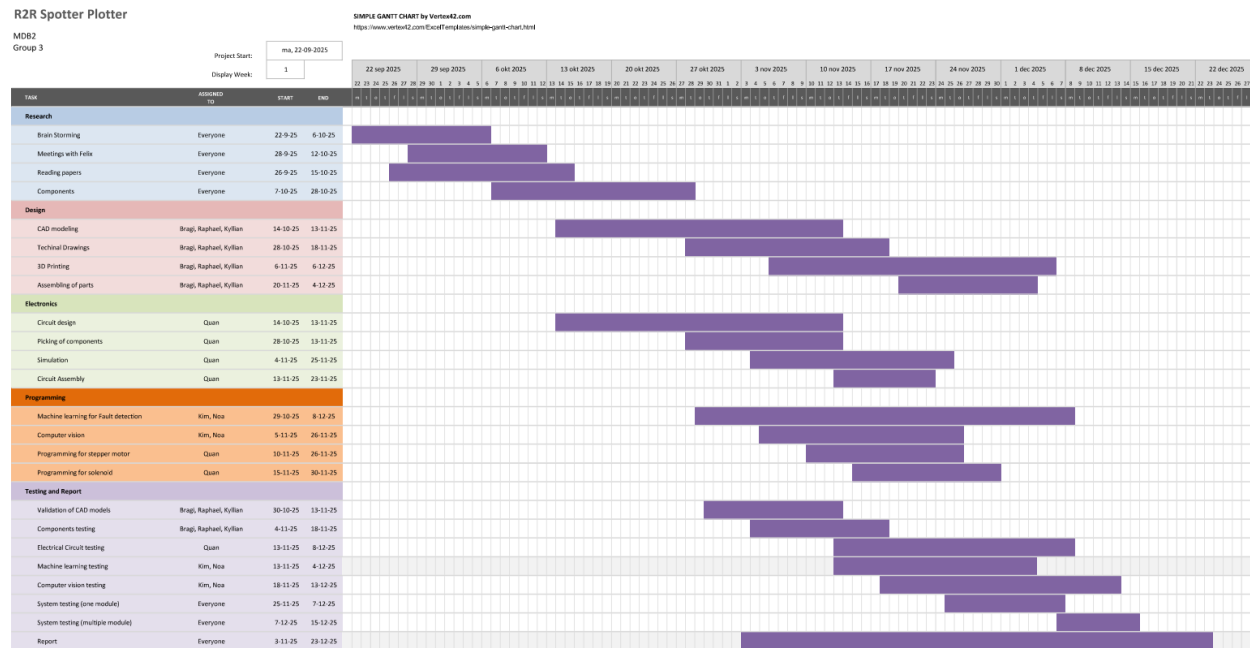


Figure 3.1: Gantt Chart

Chapter 4

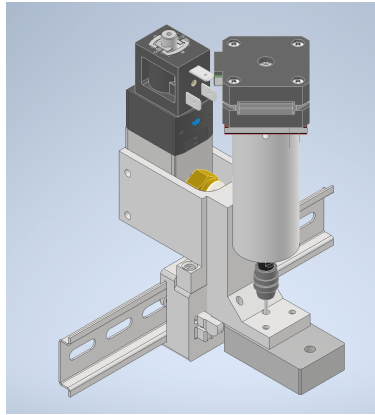
Mechanical Design

4.1 Overview

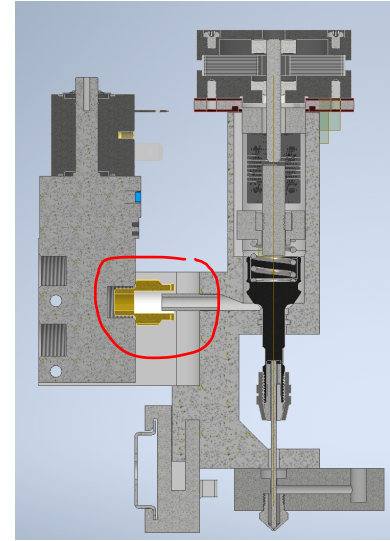
Several improvements and redevelopments had to be conducted in order to mount the new designed dispensing module on the system, and for it to work properly. Besides the new dispensing system, an attachment was made for the system to mount a DIN rail, which in turn allows for one or multiple DIN rail mountings for the dispensing modules. The R2R mechanism was redesigned so that the rolls are firmly attached to the 3D printer frame, and so that minimal slack is present when tensioning the sheet upon which the droplets are dispensed.

4.2 Droplet dispenser

The dispensing module was designed so that the needle oscillates in an upward and downward motion driven by air pressure. This motion allows the liquid to be released through the needle dispensing block. The working mechanism of the system is illustrated in the section view shown in Figure 4.1b. The dispensing module had to be designed such that it could be mounted on top of the gantry attachment without colliding with it; it was also designed to ensure easy 3D printing and simple design. Certain design choices were made to ensure the print would not break, and plenty of dimensions were precisely chosen to achieve a perfect fit. The stepper motor is connected to a shaft coupler, which is attached to an M8 bolt on the opposite side. Rotation of the M8 bolt causes the component containing an M8 nut to move vertically as it has an edge (4.2b locking it from spinning. This edge transforms the rotational motion to a vertical one with the direction of motion depending on the rotation direction of the stepper motor. This vertical movement compresses or decompresses a spring, which is used to control the maximum distance the needle can travel upward and downward. The spring provides a mechanical limit to the needle displacement due to its finite compression range.



(a) Droplet dispenser

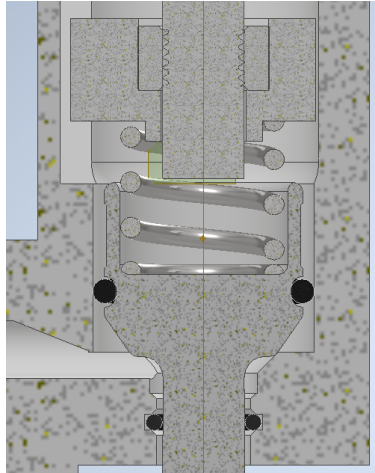


(b) Section view of dispenser

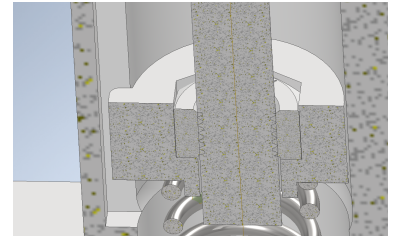
Figure 4.1: Droplet dispenser design

Several specifications were considered when selecting an appropriate solenoid valve, including the functional configuration, operating air pressure range, default (fail-safe) state, and physical size. The selected Festo solenoid valve met all of these requirements. ([2]) The valve has a 3/2 configuration which can be seen in 4.1b, which allows rapid pressure release through one inlet, one outlet, and one exhaust port. This configuration enables air to be efficiently vented from the dispensing module. The operating pressure range was chosen to match the university air supply, which provides pressures between 0 and 10 bar. The solenoid valve operates within a range of 1.5 to 8 bar, which was deemed sufficient for this application. A normally closed default state was selected to prevent air from flowing through the valve when the air supply is initially turned on, improving safety and control. Finally, the physical size of the valve was an important consideration, as the dispensing modules are designed to be stacked next to each other, requiring a compact form factor.

The black part in Figure 4.1b is the part that is moved up due to air pressure from the solenoid valve and down when the pressure is released due to the spring. To accomplish this without too much friction the fit between the PLA black part and the o-rings shown in Figure ?? has to be tight enough so that air cannot leak through which would lessen the air pressure and affect the performance and loose enough so the part can slide without too much friction which would require a strong spring. To achieve this fit, test pieces with holes with different diameters were made with to ensure that when the whole part was printed the diameters would be ideal. During the operation the top O-ring is compressed as well as expanded which creates friction between the parts whilst the small o-ring is only expanded. The friction of the small o-ring is small and will be ignored.



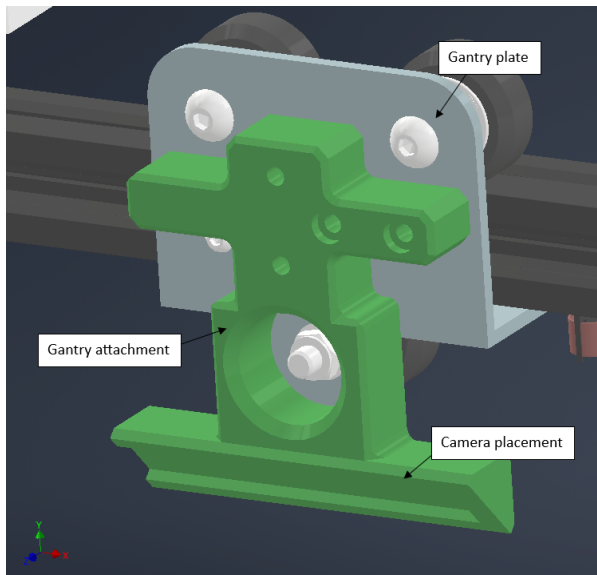
(a) Droplet dispenser



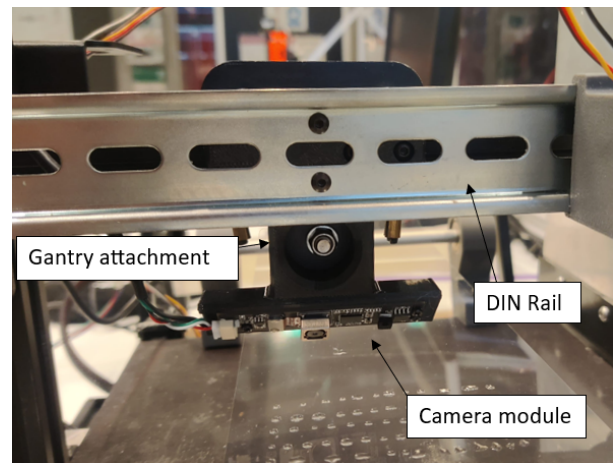
(b) Edge converting rotational to linear

4.3 Gantry attachment

Since an entirely new dispensing module was developed, a new way of attaching it to the gantry of the 3D printer had to be developed. The existing gantry attachment on the 3D printer was partially stripped, revealing a metal plate upon which a DIN rail can be mounted through the new attachment. The attachment is screwed with 2 screws to the metal plate, and the DIN rail was then also screwed in place by 2 screws on the gantry attachment, as seen in Figure 4.3. A camera placement was implemented on the underside of the gantry attachment, which will house the camera module that detects the droplets.



(a) Gantry Attachment



(b) DIN Rail screwed to Gantry attachment

Figure 4.3: Gantry attachment design

4.3.1 DIN Rail mount

The DIN rail module can slide on to the DIN rail from the side, and is then fastened to the rail with 2 pins, as can be seen in Figure 4.4, which shows the resting position of a pin anchoring itself into a slit in the DIN rail. When both pins are in their resting positions, the DIN rail mount is unable to slide horizontally.

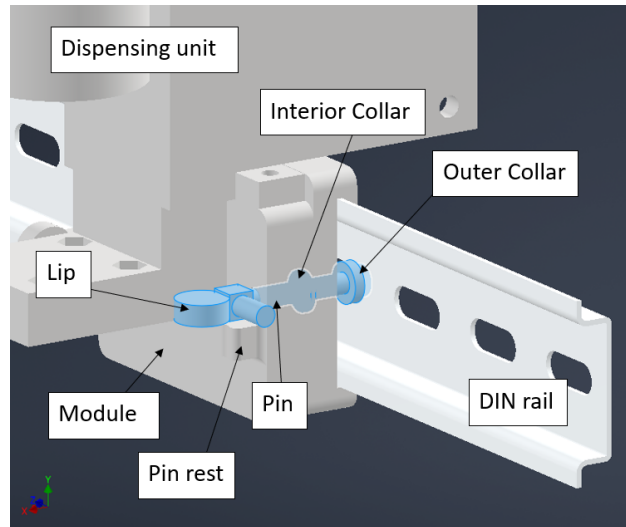


Figure 4.4: DIN Rail mount on DIN rail

The 2 pins are encased in the DIN rail module, and its movement is limited by the outer and interior collar, as seen in Figure 4.5. The movement limitations are necessary, otherwise the pin could fall out of its casing. To encase the pins together with the springs into the mount, the 3D print had to be stopped when the encasing parts were halfway printed, so that the pins together with the springs could be fit in and resume printing, finishing the encasing procedure.

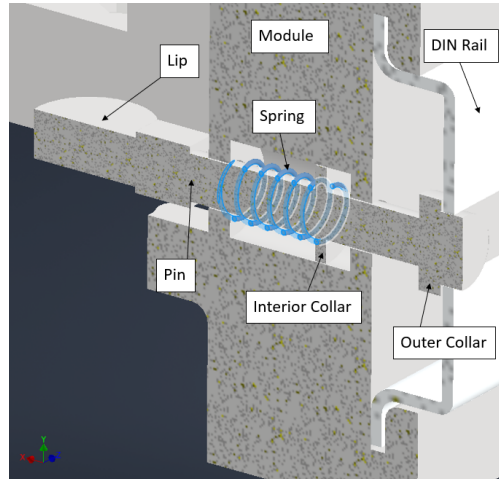


Figure 4.5: Attachment

The pin can be pulled back using the lip, compressing the spring, and twisted 90°, so that the arm rests on the pin rest, as seen Figure 4.6. This movement detaches the pin from the DIN rail slit, and when both pins are in the compressed position, the DIN rail mount can move horizontally along the DIN rail.

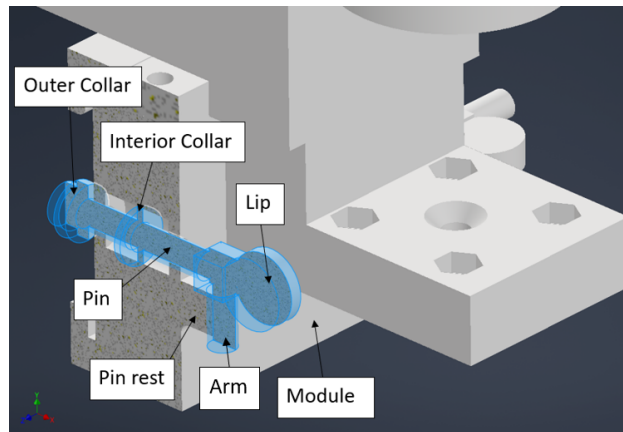


Figure 4.6: Cut view of pin

4.4 Roll-to-Roll mechanism

To enable continuous processing of a transparent substrate, a roll-to-roll (R2R) transport mechanism was integrated into the existing 3D printer platform. The system is designed as an **add-on module** and is mechanically attached to the printer frame using custom 3D printed mounting components. This approach allows the R2R mechanism to be **rigidly fixed while remaining modular and easily removable, preserving the original functionality of the printer.**

The transport system consists of two cylindrical rollers positioned upstream and downstream of the dispensing area. The transparent sheet is guided over the rollers and across the working region beneath the dispensing nozzle. Both rollers are actively driven by stepper motors to ensure controlled and synchronized motion of the substrate through the machine. By driving both rollers, the system avoids uncontrolled slack or excessive pulling forces, which are critical factors when handling thin and flexible materials.

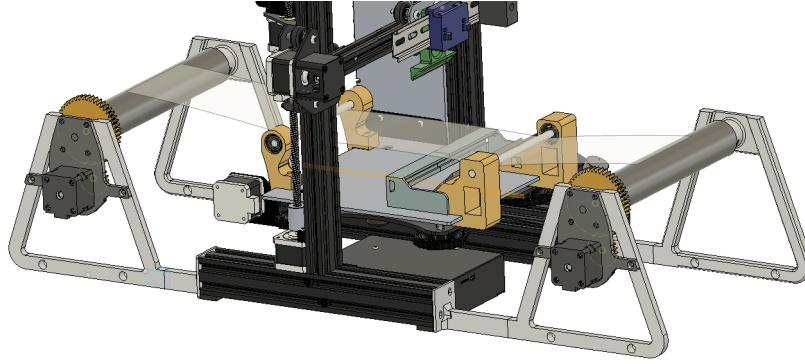


Figure 4.7: R2R mechanism

To improve torque availability and enable proper tension of the sheet, a gear reduction with a ratio of 1:2 is introduced between each stepper motor and its corresponding roller shaft. This reduction increases the effective output torque at the rollers, allowing the substrate to be tensioned sufficiently without requiring higher motor currents or larger motors. Adequate tension is essential to maintain a flat deposition surface, ensuring that droplets land on a stable and planar area without distortion caused by **sagging** or **vibrations** of the sheet.

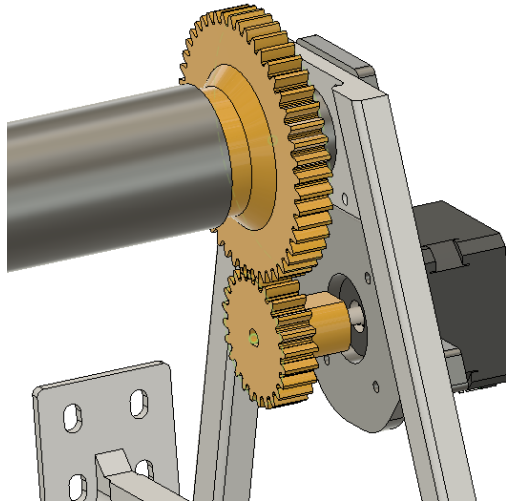


Figure 4.8: R2R gearing

The combination of rigid frame attachment, dual-motor actuation, and geared torque amplification results in a robust R2R transport mechanism capable of providing repeatable and controlled substrate motion. This mechanical stability is a prerequisite for consistent droplet placement and reliable operation of the dispensing and vision-based monitoring systems.

4.4.1 Substrate Tensioning and Grid Reference Assembly

To further improve substrate stability in the deposition area, an additional tensioning and leveling assembly was integrated between the two driven rollers. This assembly consists of an aluminium shaft supported by rolling bearings, mounted using custom 3D printed components mounted to the 3D printer bed. The transparent sheet is routed underneath this shaft, introducing a controlled deflection that increases in-plane tension while simultaneously constraining out-of-plane motion.

The aluminium shaft provides a rigid and dimensionally stable reference, while the bearings minimize friction to allow smooth substrate motion during roll-to-roll operation. By applying a consistent downward force on the sheet, the assembly helps eliminate sagging and local height variations, ensuring that droplets are deposited onto a flat and tensioned surface. This is particularly important for reliable droplet formation and placement, as variations in substrate height directly affect droplet impact behavior and apparent footprint size.

In addition to its mechanical function, this assembly also serves as the mounting location for the reference grid used in droplet detection and calibration. The grid is printed on a sheet of paper with known geometry and embedded directly into a 3D printed frame. To achieve accurate and repeatable positioning, the 3D print is paused at a predefined layer height, the printed grid is manually inserted into the partially printed frame, and the print is then resumed. This process permanently encapsulates the grid within the frame, fixing its position relative to the machine coordinate system.

By combining substrate tensioning, leveling, and grid referencing into a single structural element, the assembly reduces system complexity while improving measurement repeatability. The fixed spatial relationship between the grid, the substrate, and the camera enables consistent calibration and robust droplet detection throughout operation.

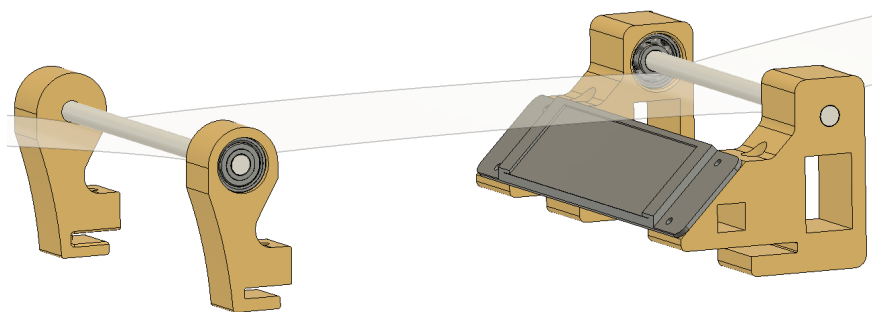


Figure 4.9: R2R Tensioning brackets

4.5 Needle Dispensing Block Design and Development

The needle dispensing block is a critical mechanical component of the R2R Spotter Plotter system. Its primary function is to transport liquid from the syringe through internal flow channels and deliver it to the substrate via a dispensing needle. In addition to guiding the fluid, the block provides precise mechanical support for the needle and serves as the interface between the fluid delivery system and the dispensing mechanism. The design ensures stable needle positioning and reliable fluid flow during operation.

4.5.1 Design Concept

The initial design of the needle dispensing block is shown in Figures 4.10 This design was made by a previous person and was used as a starting point for the fluid dispensing concept [1].

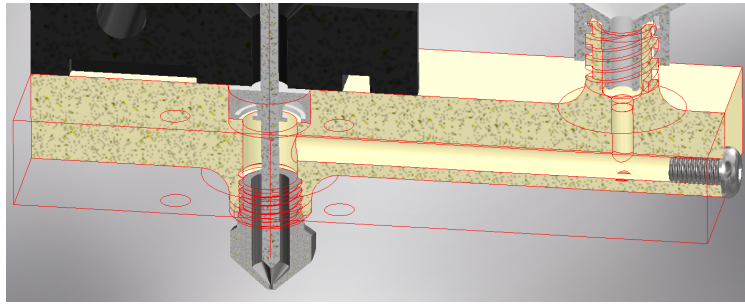


Figure 4.10: Initial needle dispensing block design

Since the idea is to make the part out of metal, some parts of the design might be difficult to manufacture. To make the part easier to manufacture and more suitable for metal production, a new version of the needle dispensing block was developed. The updated design is shown in Figures 4.11 4.12.

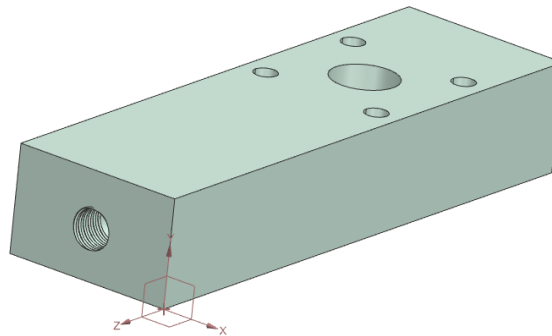


Figure 4.11: Revised needle dispensing block design

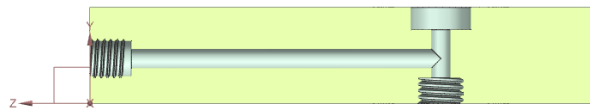


Figure 4.12: Initial needle dispensing block design section view

In the new design, the internal geometry was simplified. Straight cylindrical holes were used instead of an outside thread for the needle, and instead of the liquid having to take a 90-degree turn. This also means one less hole to drill, which makes the part easier to machine using standard drilling. A thread is then made for the Luer lock to be secured in.

Even though the geometry was simplified, the function of the block remains the same. Liquid flows now from the side inlet instead of from above and out through the needle in the same way as in the original design.

An updated version of the dispensing block design is shown in Figure 4.13. Compared to the previous iteration, the overall length of the component has been reduced. This modification was implemented after

it was determined that the additional length did not contribute to either structural integrity or functional performance. By shortening the component, the design is simplified while preserving its intended mechanical functionality. The part was manufactured using PETG, selected for its improved water resistance compared to standard PLA. Additional wall thickness was incorporated around critical regions to enhance sealing and durability. While these design features improve suitability for polymer-based fabrication, they increase the complexity of manufacturing the component using conventional metal machining techniques. With ongoing advancements in additive manufacturing, the component could potentially be produced using metal additive processes. Alternatively, more cost-effective fabrication methods such as SLA or SLS printing may offer improved surface quality and sealing performance while maintaining sufficient mechanical strength. The reduced size also improves material efficiency and manufacturability, making the part more suitable for rapid prototyping and integration into the overall system without negatively affecting performance.

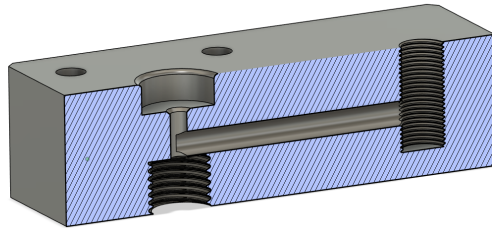


Figure 4.13: Updated Block design

4.5.2 Design Rationale

The new needle dispensing block was designed with a focus on simplicity, robustness, and ease of manufacturing. By adapting the geometry for metal machining while keeping the same functional layout, the design is better suited for prototyping and integration into the R2R Spotter Plotter system. In this project, the part was 3D printed since it is a prototype, and if changes were made, it would not be needed to manufacture the part every time, but in future applications, it is recommended to make the part out of metal.

4.5.3 Luer Lock Fluid Connection

To connect the needle dispensing block to the syringe system Luer lock connectors were made. Luer lock were chosen because they provide a secure leak-resistant connection and are commonly used in fluid applications.

A modified Luer lock connector, shown in Figure 4.14, is mounted on the side of the needle dispensing block. This connector allows flexible tubing to be attached directly to the block, providing the fluid inlet. The internal thread in the block secures the Luer lock fitting and ensures a stable connection during operation.

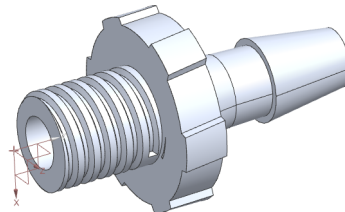


Figure 4.14: Luer Lock Connecting to block

The tubing connects the dispensing block to a second Luer lock connector, shown in Figure 4.15, which is attached to the syringe. When pressure is applied to the syringe, liquid is pushed through the tubing into the dispensing block and finally out through the needle.

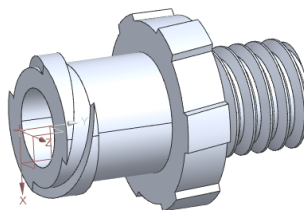


Figure 4.15: Luer Lock Connecting to block

This setup allows the syringe to be easily connected and removed while maintaining reliable fluid flow. Using standard Luer lock components also simplifies assembly and makes the system compatible with commonly available syringes and tubing.

4.5.4 Gear-Based Syringe Actuation Mechanism

To push liquid out of the syringe in a controlled manner, a mechanical actuation system based on gears was designed. The purpose of this system was to convert the rotational motion of a stepper motor into linear motion that pushes the syringe plunger.

The mechanism is shown in Figures 4.16 and 4.17. A stepper motor is connected to a worm gear, which drives a larger spur gear. This spur gear is mechanically linked to a second gear that engages with a linear rack. As the stepper motor rotates, the worm gear turns the spur gears, causing the rack to move linearly and push the syringe plunger forward.

The worm gear was chosen to provide a high gear reduction, allowing the stepper motor to generate sufficient force to push the syringe plunger. The use of a rack and pinion mechanism converts this rotational motion into linear displacement. However, it was found that this mechanism does not maintain a constant pressure in the fluid system. When the dispensing needle moves, changes in flow resistance occur, which lead to pressure drops in the system. Since the gear-based mechanism controls position rather than pressure, it is not able to compensate for these pressure variations. As a result, the liquid flow rate becomes inconsistent, which limits the precision of the dispensing process. Although the mechanism successfully demonstrates a working concept for mechanically pushing the syringe plungers, it highlights the challenge of maintaining stable pressure using a purely position-controlled mechanical actuation system.

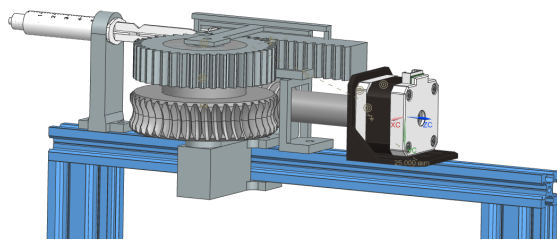


Figure 4.16: Gear-based syringe actuation CAD model

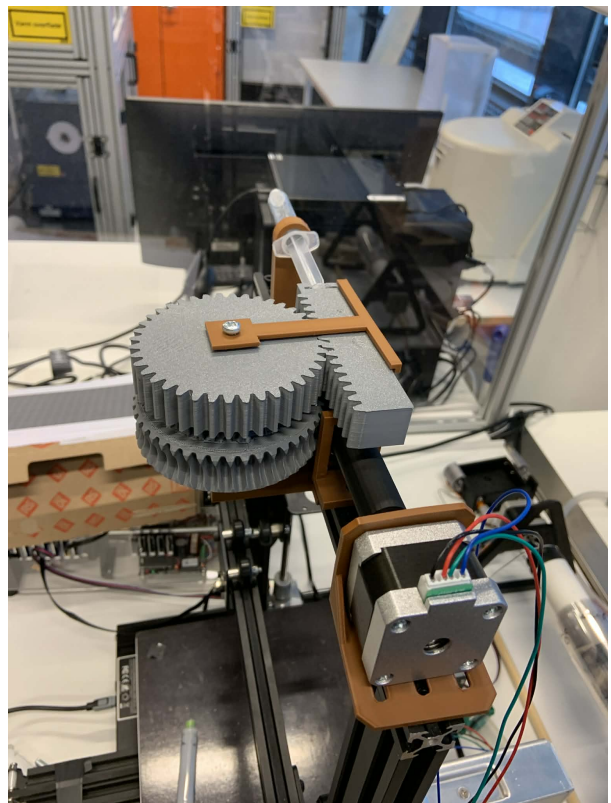


Figure 4.17: Gear-based syringe actuation printed prototype

Chapter 5

Electronic design

This section presents the electronic control board designed for the drop-dispensing system. The main goal of this board is to control two key elements: first, the stepper motor used to adjust the droplet size; and second, the solenoid valve used to dispense the drop. This board provides a more affordable solution compared to the previous approach, which relied on an external pulse generator and a piezoelectric actuator with a dedicated PDu150CL driver, significantly increasing the system cost. In addition, the new solution offers a more flexible platform that can be used with multiple dispensing modules.

5.1 System Architecture

The electronic system is powered from a main 24V DC power source. The power was divided into two: powering the motors/solenoid, and powering the controller at 12V. This 5V was generated through a linear step-down regulator, from 24V to 12V by a LM317 and from 12V to 5V by L7805. This 5V power line is the power for the step motor driver and the multiplexer. The overall system can be seen in Figure 5.1.

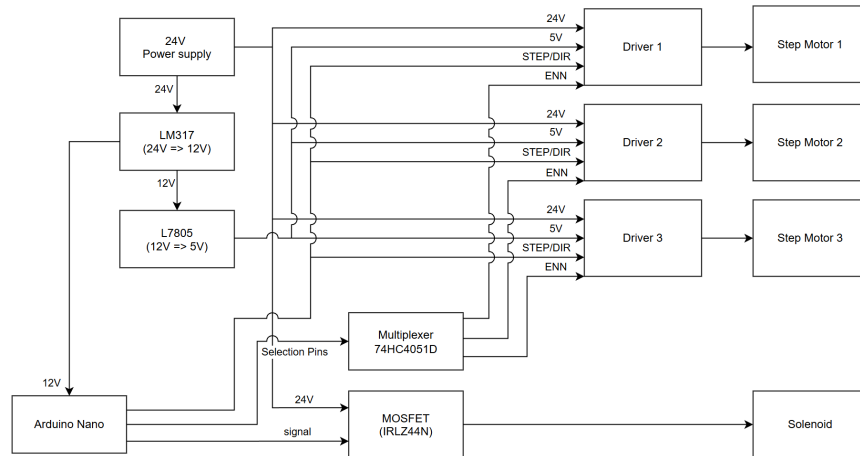


Figure 5.1: Overall electric board design

The Arduino Nano is responsible for generating deterministic STEP and DIR signals for controlling motor motion, as well as a digital control signal for switching the solenoid valve state.

5.2 Component Selection

The component choices were made to satisfy voltage/current requirements, enable accurate timing control, and still able to keep the cost low and able to scale.

- Microcontroller (Arduino nano)
- Enable channel (74HC4051D)
- Stepper motor drivers (TMC2225)
- Stepper motors (NEMA17)
- Solenoid valve
- Power regulation (LM317 and L7805)
- Solenoid switch (IRLZ44N)

5.3 Circuit Design

5.3.1 Power supply

The main power supply was divided into two distribution paths. The first path delivers power directly to the motors and the solenoid, which are connected in parallel. The second path provides regulated logic-level voltages through a linear regulation stage. The incoming 24V supply is first reduced to 12V using an LM317 voltage regulator, and subsequently stepped down to 5V using an L7805 regulator. The 5V supply powers the Arduino Nano and the TMC2225 motor driver logic circuits. In addition, the 5V rail serves as the reference voltage for the pull-up resistors used in the multiplexer circuitry.

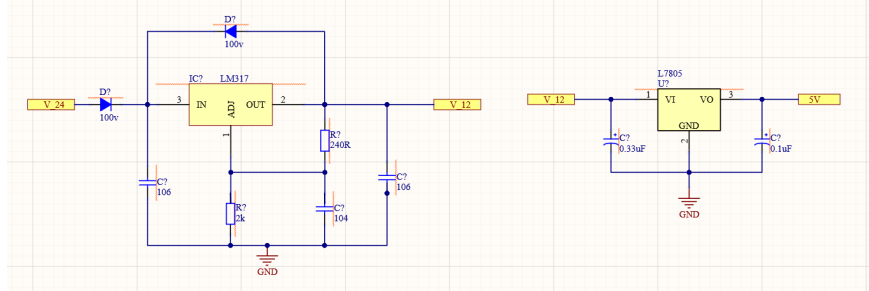


Figure 5.2: Power supply circuit

5.3.2 Stepper motor driver

Each stepper motor is driven using a TMC2225 driver operating in standalone STEP/DIR mode, where the Arduino Nano will generate STEP pulse and DIR signal to command motor velocity and direction. The microcontroller can choose which driver to work with through the enable pin (ENN). The ENN pin is active-low, so in the implemented electric board, each ENN lines are connected to a pull-up resistor. With the MS1 and MS2 pins connected to ground, the motor runs with 1/4 microstepping according to the TMC2225 MS pin table [3]. With the step motor is 200 STEP per revolution [4], result in the command revolution becomes:

$$\text{Steps/rev} = 200 \times 4 = 800$$

5.3.3 Multiplexer

22

5.3.4 Solenoid drive stage

The solenoid valve is powered from 24V main source and controlled using a MOSFET low-side switch. As the coil is inductive, a flyback suppression path is required to limit the turn-off voltage spike.

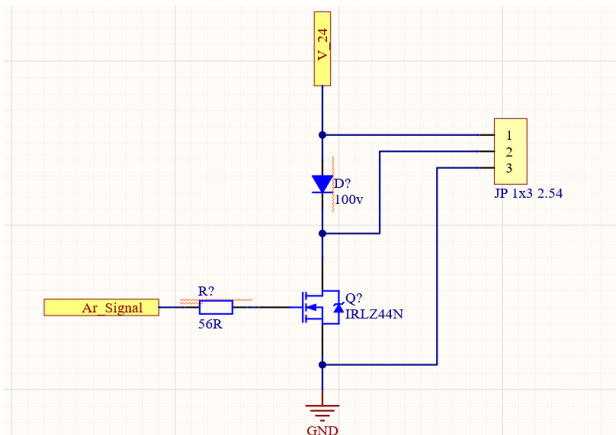


Figure 5.5: Solenoid circuit design

5.4 Main Board

In Figure 5.6, the board was divided into four functional regions: TMC2225 driver modules, solenoid and its MOSFET switch, Arduino Nano controller, and the power regulation section. The power distribution uses a shared 24V with parallel branches to each driver and the solenoid, while the LM317 to L7805 chain generates 12V and 5V supply for Arduino and driver logic supply.

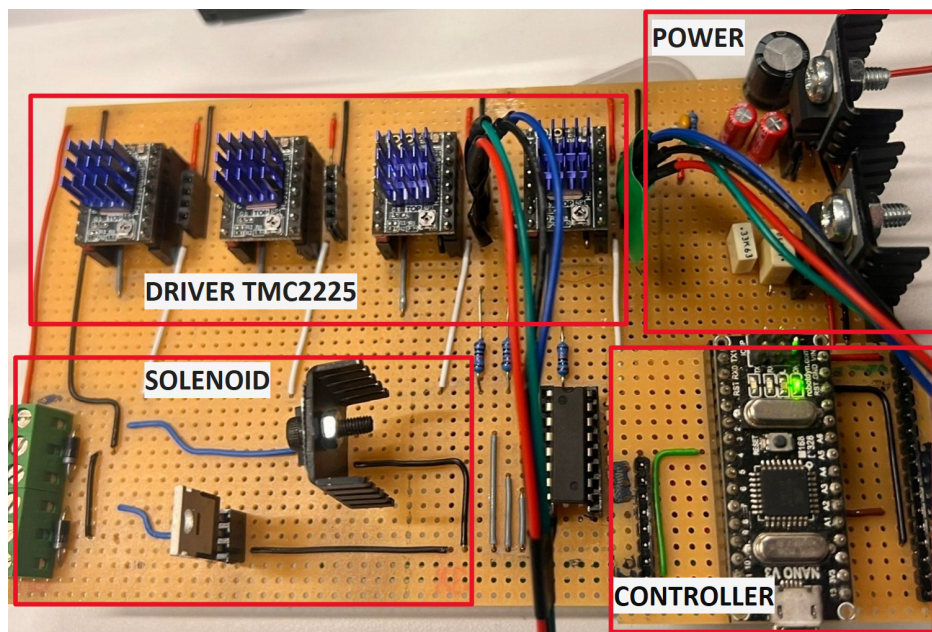


Figure 5.6: Main Board design

Chapter 6

Programming and Software Implementation

6.1 Overview

The programming of the system is divided into two primary software components:

1. **PC-side control and vision software**, implemented in Python
2. **Embedded firmware for motor control**, implemented in Arduino IDE (on an Arduino Nano)

The PC-side application acts as the central coordinator, responsible for user interaction through a Graphical User Interface (GUI) seen in Figure 6.1, motion sequencing of dispensing units, computer vision processing, and synchronization between subsystems. The embedded Arduino firmware is responsible for controlling the R2R mechanism, dispensing mechanism, and feedback control on droplets.

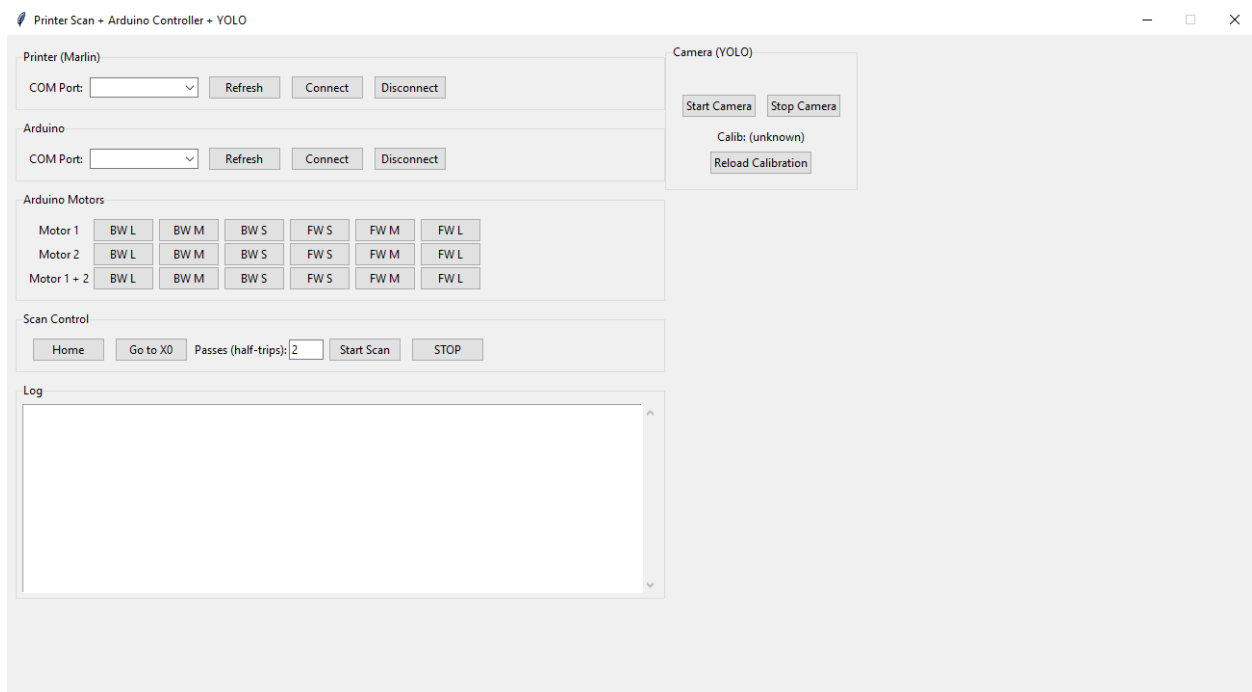


Figure 6.1: Graphical User Interface

Communication between the PC to the 3D printer and Arduino, as well as camera feed are all done through independent serial connections, requiring 3 available ports on the PC. The 3D printer is responsible for moving the dispensing units across the sheet to create the grid of droplets. The sheet is moved through the control of the Arduino, as well as controlling spring compression on the dispensing units.

6.2 Software Architecture

6.2.1 System Architecture

The software architecture follows a centralized control model in which the PC application plans all system behavior. The 3D printer is used to generate precise linear motion of the dispensing units across the R2R substrate, while the Arduino controls auxiliary mechanisms related to the R2R system and pneumatic dispensing actuation. The PC application coordinates these subsystems by sending motion commands to the printer, triggering auxiliary motor movements on the Arduino at defined points in the scan sequence, and processing visual feedback from the camera system via AI detection. Each subsystem operates independently but is synchronized through explicit command acknowledgments to ensure deterministic and repeatable operation.

6.2.2 PC-side Application Architecture

The PC application is implemented as a single Python program structured around a main GUI class (CombinedGUI) and a dedicated camera processing thread (YOLOCamera). The application integrates four primary responsibilities:

- Graphical User Interface
- Serial Communication with 3D printer
- Serial Communication with Arduino
- Real-time computer vision and measurement

Threading and message queues are used extensively to maintain responsiveness and prevent blocking behavior in the GUI event loop. Serial receive threads continuously process incoming data from both the printer and Arduino, while synchronized queues are used to ensure that motion commands and acknowledgements are handled in a controlled manner.

6.2.3 Graphical User Interface Architecture

The GUI is implemented using the tkinter library and creates organized sections for the user. These sections include:

- Printer connection
- Arduino connection and motor control
- Scan control and motion sequencing
- Logging of commands
- Camera display and droplet detection

This layout improves usability and allows each subsystem to be operated independently, while remaining part of a unified control interface.

6.3 Software Functionality and User Interaction

This section describes the operational behavior of the software from the user's perspective and explains how the system responds to user input via the GUI.

6.3.1 Manual Arduino Motor Control

The **Arduino Motors** control section provides manual actuation of the R2R mechanism, allowing each motor to be moved independently. This is intended for use when setting up the sheet, enabling an easy way to roll up a sheet on the R2R. Additionally, it is also for tensioning the sheet by getting a motor to pull on either side, so it maintains a flat horizontal surface for the droplets to land.

6.3.2 Homing and Referencing

The **Home** function initiates the common homing function a 3D printer does before starting to print. It finds the coordinate origin of the used axes and is fundamental to ensure known position of the dispensing units. For this project, the Y axis is not used, as the R2R mechanism is controlled by the Arduino instead.

The **Home** function starts with finding X0 with a limit switch, then finding Z0 with a BL-touch contacting the sheet. Afterwards, it moves Z to a determined height for dispensing droplets.

6.3.3 Manual Position Reset (Go to X0)

The **Go to X0** moves the printer dispensing head to the zero position on the X axis using absolute positioning. This allows the user to move the head out of the way, if there is a need to get better access to the sheet.

6.3.4 Automated Scan Operation

When the **Start scan** function is activated, the system executes an automated sequence consisting of a predefined number of half trips along the X axis, chosen by the user. Each half trip corresponds to one row on the sheet.

Before scanning begins, motion parameters such as speed, acceleration, and jerk are configured to ensure controlled and repeatable motion. The head then reset to the starting position at X0, which is not counted as part of the scan.

During each half-trip, the printer moves from one side to another. When it gets to either end of the sides of the sheet, the Arduino is triggered to move the sheet, such that the head can dispense on another row. This auxiliary action is synchronized with the printer motion and only executed after confirmation that the printer has completed its current movement.

6.3.5 Emergency Stop

The **Stop** function provides an immediate interruption of system operation. When activated, a stop command is sent to the printer, halting all ongoing motion and terminating the scan loop. This functionality allows the user to safely interrupt operations in the event of an error or unexpected behavior.

6.3.6 Logging

The GUI has a log window useful for development and debugging, by showing the commands sent and received by the PC from the printer and Arduino. As this prototype is still not fully functioning, the log window remains for further development.

6.4 Motion Control and System Synchronization

Motion control is achieved through G-code commands issued to the printer controller over a serial connection. Fixed feedrate (speed), acceleration, and jerk parameters are used to ensure predictable mechanical behavior and to prevent unsafe operating conditions.

Synchronization between printer and Arduino subsystems is achieved through explicit acknowledgement-based communication. Printer motion is considered complete only after receiving confirmation following a blocking motion command, while the Arduino motor actions are confirmed through a completion response. This approach avoids time-based delays and ensures that each mechanical action completes fully before the next one begins.

6.5 Embedded Firmware (Arduino)

The Arduino firmware is responsible for low-level control of dispensing mechanisms and auxiliary motors. It receives simple, structured commands from the PC over a serial connection, executes the requested motor movements using deterministic pulse generation, and reports completion back to the PC.

By delegating timing-critical motor control to the microcontroller, the system avoids operating system scheduling delays and ensures consistent mechanical behavior. All high-level sequencing and decision-making remain on the PC side.

Chapter 7

Machine Learning & AI Droplet Detection

7.1 Overview

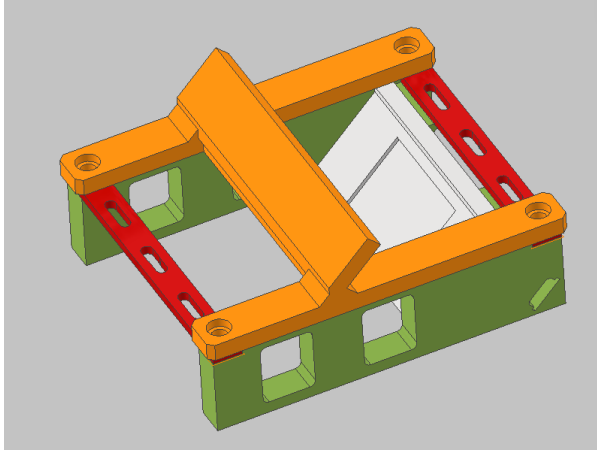
To implement the AI-based droplet detection, a small-scale testing setup was first developed to verify the feasibility of the approach on a limited dataset. This preliminary setup allowed for controlled data collection, model training, and evaluation of the detection performance under simplified conditions. The results demonstrated that the model was able to successfully learn and identify droplets with sufficient accuracy, thereby validating the underlying concept. Based on the positive outcomes of the training and testing phase, it was concluded that the approach is viable and robust enough to be implemented on the full-scale roll-to-roll (R2R) system.

7.2 Prototype test setup and detection method

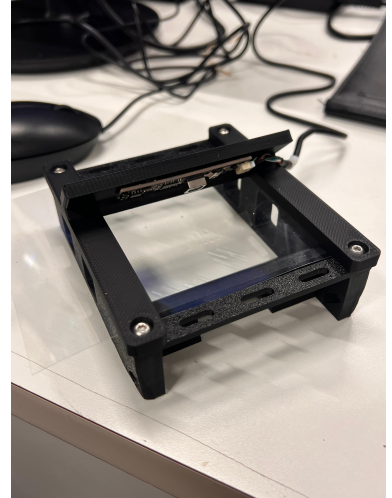
7.2.1 Test setup

Creating the test setup allowed experimental validation of the droplet detection system without relying on the full R2R process being operational. The test setup, seen below in Figure 7.1, was designed to replicate the relevant conditions of the R2R Spotter Plotter closely. The camera was rigidly mounted in a fixed position, ensuring a stable and repeatable imaging configuration. Additionally, a transparent sheet - corresponding to the substrate used in the R2R process - was securely positioned beneath the camera. This sheet served as the surface onto which droplets would normally be deposited during operation. By isolating the vision system from the remaining hardware dependencies, this setup enabled controlled testing, data collection, and performance evaluation of the droplet detection algorithm under representative geometric and optical conditions.

The test setup was fully fabricated using 3D printing, allowing for rapid prototyping, easy modification, and precise alignment of components. This modular design enabled efficient iteration during the development and testing phases. The camera was mounted at a 45° angle relative to the transparent substrate, replicating the intended mounting configuration on the R2R machine. This orientation was selected to accurately represent the operational working range and viewing geometry of the final system. A grid template was integrated into a dedicated 3D-printed holder, ensuring stable positioning and repeatable alignment throughout the experiments. Consistent with the camera configuration, the grid was also mounted at a 45° angle, maintaining geometric alignment between the imaging plane and the reference pattern. The grid holder was designed to place the template at a precise distance corresponding to the camera's focal range, resulting in a sharp and high-contrast grid image that improves detection reliability. To further enhance image quality, an LED strip was incorporated to provide controlled illumination of the grid. This additional lighting improves the visibility of grid distortions caused by the droplets, thereby facilitating more robust and consistent droplet detection under varying ambient lighting conditions.



(a) Test setup CAD



(b) Actual test setup

Figure 7.1: Test setup

7.3 Detection Method

7.3.1 Grid-Based Droplet Detection Using Optical Distortion

The droplet detection approach employed in this project is based on exploiting optical distortion effects induced by liquid droplets placed on a transparent substrate above a reference grid. This method was selected to address the inherent challenge of detecting transparent liquids on transparent surfaces, where conventional intensity or color-based computer vision techniques typically fail due to insufficient contrast.

A grid pattern with known and uniform spacing is positioned beneath the transparent sheet and within the camera's field of view. In the absence of droplets, the grid appears as a regular and approximately linear structure in the captured image, providing a stable geometric reference. When a droplet is present, it locally distorts the appearance of the grid due to refraction at the liquid-air and liquid-substrate interfaces. These distortions happen from changes in the optical path of light rays caused by differences in refractive indices, in accordance with principles of geometric optics [5].

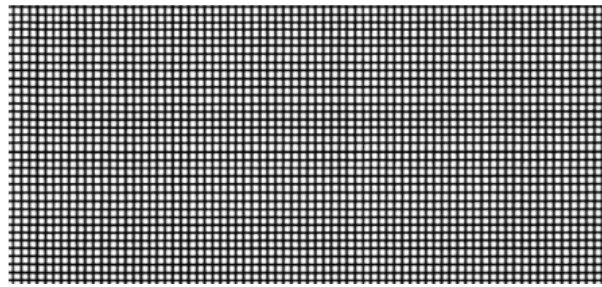


Figure 7.2: Grid pattern

From a physical standpoint, each droplet behaves as a weak refractive lens with a curved surface. Incoming light rays passing through the droplet are bent according to Snell's law, producing spatially varying displacement and deformation of the underlying grid lines [5]. The magnitude and spatial extent of this distortion are directly related to the droplet's geometry, including its diameter, curvature, and volume. Larger droplets introduce stronger refraction and affect a wider region of the grid, while smaller droplets produce more localized distortions. As a result, the distorted grid region provides an indirect but reliable representation of droplet size and shape.

Due to the fixed camera mounting at a 45° angle relative to the substrate, perspective projection effects further influence the appearance of both the grid and droplet-induced distortions. Under this viewing geometry, circular droplets do not appear circular in the image plane but instead generate elliptical or elongated distortion patterns. Rather than compensating for these effects through explicit geometric rectification or camera calibration, the detection system is trained directly on images captured under the final operating conditions. This allows the model to implicitly learn the mapping between the physical geometry of the droplets and their distorted image representation, an approach commonly adopted in machine vision systems that rely on structured visual references [6].

Droplet detection is performed by identifying regions where the local grid structure deviates from its expected regularity. These regions are enclosed using bounding boxes and classified as normal or faulty droplets on the basis of learned distortion characteristics. Normal droplets typically generate compact and symmetric distortion fields, whereas faulty droplets, such as merged droplets, elongated spills, or irregular deposits, produce asymmetric, extended, or overlapping distortion patterns.

To improve computational efficiency and robustness, detection is restricted to a predefined region of interest aligned with the expected droplet row. This reflects the structured nature of the R2R dispensing process and reduces the influence of irrelevant grid regions. Overall, the grid-based distortion approach transforms a low-contrast optical detection problem into a geometrically structured one grounded in physical optics, enabling reliable droplet detection and **fault classification** under realistic industrial conditions. An example of the output of the detection on the test setup can be seen below in Figure 7.3.

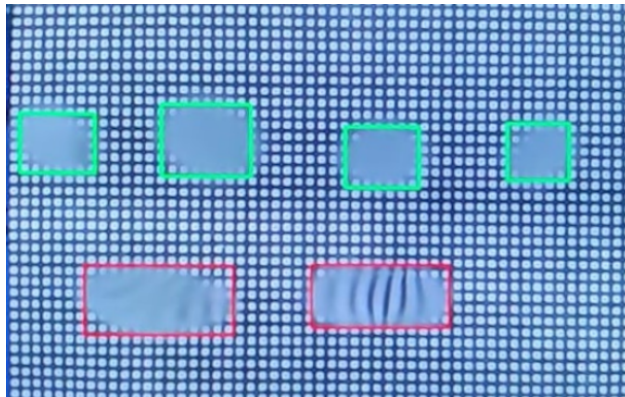


Figure 7.3: Display of detection on prototype test setup

7.4 Data Collection and Processing

7.4.1 Data Collection

The dataset used for training and evaluation was collected using the experimental test setup described in the previous section. Images were captured with a fixed camera mounted at a 45° angle relative to the transparent substrate, replicating the intended viewing geometry of the R2R dispensing system. A grid pattern with known spacing was placed beneath the transparent sheet and served as a visual reference for droplet-induced optical distortion. Data acquisition focused on capturing both correctly formed droplets and representative fault cases. Normal droplets correspond to well-dispersed, isolated droplets with regular geometry, while **faulty** droplets include irregularly shaped droplets, elongated deposits, merged droplets, and spill-like structures. This ensured that the dataset reflects realistic process variations and potential failure modes encountered during operation.

Images were recorded under controlled lighting conditions using an LED illumination source to enhance grid visibility and ensure consistent contrast. While ambient lighting variations were minimized, small natural variations were intentionally retained to improve model robustness. The camera position, focus, exposure, and field of view were kept constant throughout data collection to maintain a fixed geometric relationship between the grid, substrate, and imaging system. To reduce computational complexity and better reflect the

structured nature of the dispensing process, only the relevant portion of the image containing a single row of droplets was retained for further processing. This region of interest corresponds to the expected droplet deposition path in the R2R system and allows for more efficient training and inference.

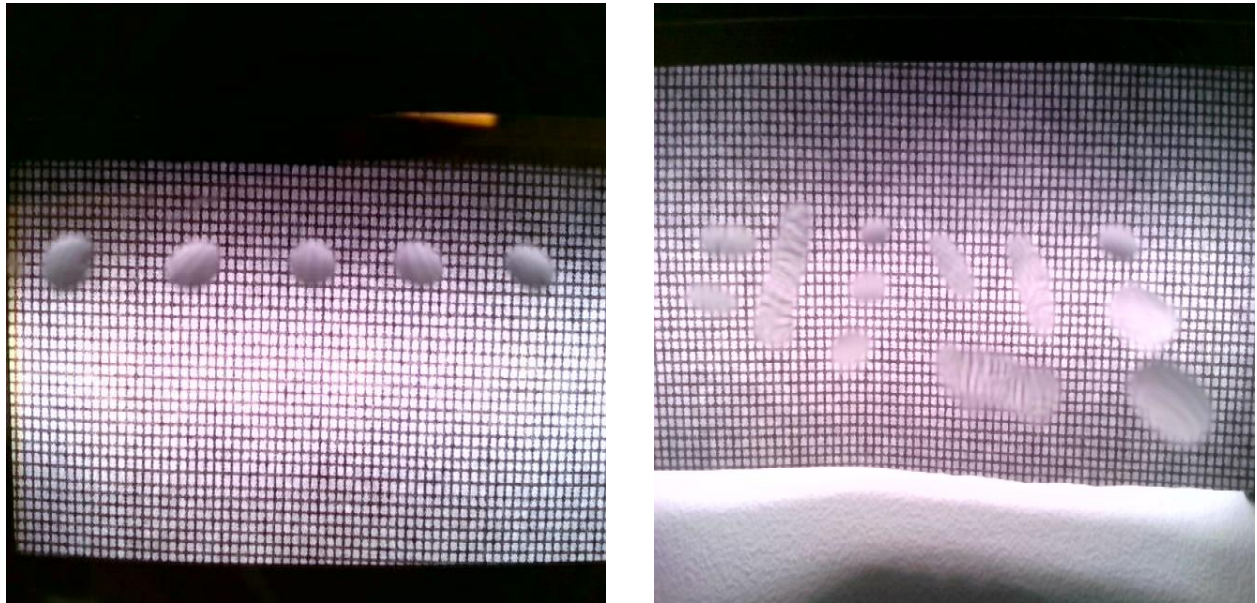


Figure 7.4: Images taken for training

7.4.2 Data Processing and Preprocessing

Following data collection, all images were manually annotated using a dedicated labeling tool. Each droplet was labeled individually and classified into one of two categories: normal droplets and faulty droplets. Labels were assigned based on the visual characteristics of the grid distortion induced by each droplet. Compact and symmetric distortion patterns were classified as normal, while asymmetric, elongated, overlapping, or irregular distortion patterns were classified as faulty.

Segmentation-based annotations were used to label individual droplets within the images. Rather than relying on simple bounding boxes, pixel-wise segmentation masks were employed to more accurately capture the spatial extent and shape of the droplet-induced grid distortions. This approach is particularly well-suited to the present application, as droplet boundaries are not sharply defined due to optical refraction and transparency, making precise localization based solely on edges unreliable. The use of segmentation labels enables the model to learn fine-grained shape and distortion characteristics associated with normal and faulty droplets. Although segmentation-based annotation is more time-consuming than bounding box labeling, it provides richer spatial information and improves robustness in cases involving irregular, elongated, or merged droplets. For inference and deployment, the segmentation output can be reduced to bounding boxes or class-level detections to meet real-time and embedded performance requirements.

Prior to training, the dataset was divided into training, validation, and test subsets to enable unbiased performance evaluation. Standard preprocessing steps were applied, including resizing images to a fixed input resolution compatible with the detection model. In addition, data augmentation techniques such as small rotations, brightness adjustments, and minor scaling were applied to improve generalization performance while preserving the physical realism of the droplet-induced distortions.

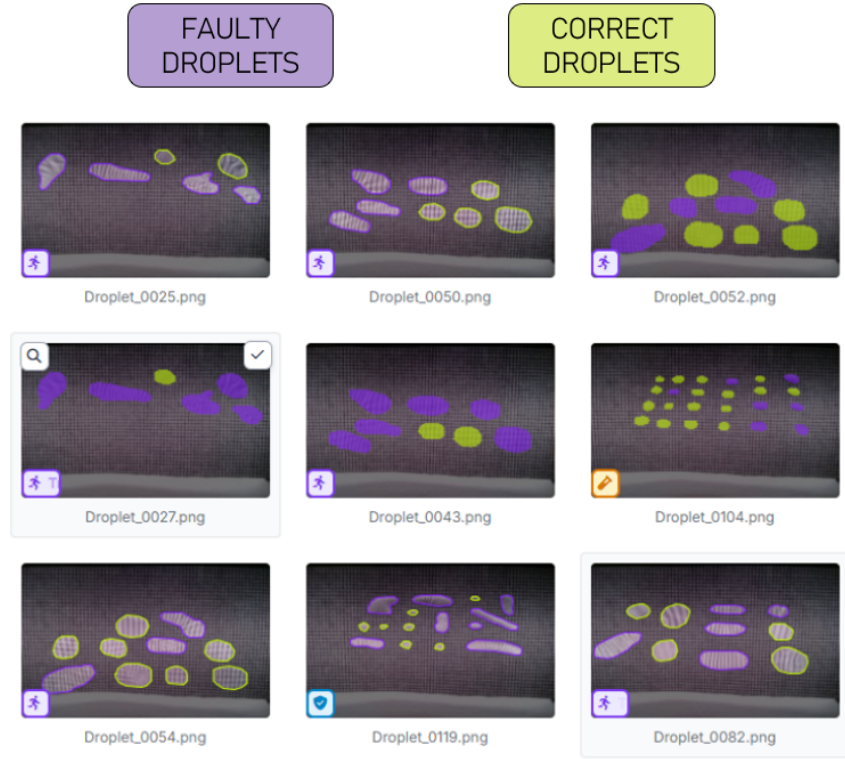


Figure 7.5: Labeled data using Roboflow

7.5 Model Architecture and Training

7.5.1 Model Architecture

For the task of detecting and classifying liquid droplets on a transparent substrate, a YOLO-based deep learning model was selected. YOLO (You Only Look Once) models are single-stage object detectors that perform localization and classification in a single forward pass, making them particularly suitable for applications with real-time or near real-time constraints.

YOLOv8 was chosen due to its improved detection accuracy, efficient architecture, and native support for both object detection and segmentation tasks. In this project, the segmentation variant of YOLOv8 was employed to leverage pixel-level annotations during training. This enables the model to learn fine-grained spatial and shape-related features associated with droplet-induced grid distortions, which are critical for distinguishing between normal and faulty droplets.

The model architecture consists of a convolutional backbone responsible for feature extraction, followed by a neck component that aggregates multi-scale features, and a task-specific head that outputs class predictions and segmentation masks. This multi-scale design is well-suited for detecting droplets of varying apparent size and distortion strength caused by perspective projection and refraction effects.

7.5.2 Training Setup

Training was performed locally on a GPU-enabled system using the Ultralytics YOLOv8 framework. Local training enabled faster experimentation, direct control over hyperparameters, and realistic benchmarking prior to embedded deployment. The input image resolution was selected to balance detection accuracy with computational efficiency, taking into account the constraints of future deployment on embedded hardware.

No geometric rectification or perspective correction was applied prior to training. Instead, the model was

trained directly on images captured under the final operating geometry, allowing it to implicitly learn the effects of refraction and perspective projection present in the system.

7.5.3 Training Procedure and Evaluation Metrics

The training objective was formulated as a dual-class segmentation problem, where the model predicts both the class label and spatial extent of each detected droplet. Optimization was performed using stochastic gradient descent with adaptive learning rate scheduling, as provided by the YOLOv8 training framework.

Model performance was evaluated using standard object detection and segmentation metrics. Precision and recall were used to assess detection reliability, where precision reflects the model’s ability to avoid false positives and recall indicates its ability to correctly identify all droplets present in the image. Mean Average Precision (mAP) at an Intersection-over-Union (IoU) threshold of 0.5 was used as the primary accuracy metric, representing overall detection performance under moderate localization requirements. In addition, mAP averaged over IoU thresholds from 0.5 to 0.95 was considered to evaluate localization and segmentation quality under stricter conditions, making it more sensitive to boundary accuracy and shape estimation. The evaluation results are shown in Figure 7.6.

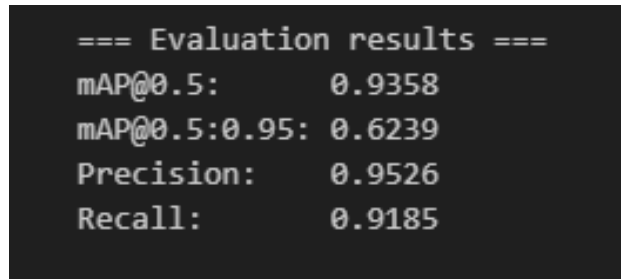


Figure 7.6: Evaluation metrics

These metrics provide a comprehensive evaluation of the model’s ability to accurately detect droplets, distinguish between normal and faulty cases, and localize distortion regions under realistic imaging conditions.

7.6 AI-Based Droplet Detection on the R2R System

After validation on the prototype setup, the trained deep learning model was integrated into the full R2R system. The pretrained YOLO-based segmentation model was deployed on a dedicated computer and executed in real time through a custom GUI. The GUI manages camera acquisition, model inference, and visualization of detection results, enabling continuous monitoring of droplet deposition during operation.

Once activated, the system performs droplet detection directly on the live camera stream. Consequently, raw pixel-based measurements cannot be directly converted into physical droplet dimensions without geometric correction.

To enable reliable droplet size estimation, a dedicated calibration procedure is required. This calibration establishes a mapping between the distorted image space and a planar, metric reference frame aligned with the substrate. The mapping is computed using a planar homography, which allows perspective effects introduced by the angled camera to be compensated prior to measurement [7].

7.6.1 Camera Calibration and Homography-Based Measurement

To compensate for the perspective distortion introduced by the angled camera, a homography-based calibration approach is employed. Under the assumption that droplets are deposited on a planar surface and that the reference grid lies in the same plane as the camera, a projective transformation can be used to map image coordinates to a rectified top-down view of the substrate [7]. A reference grid with known geometry is embedded into a rigid 3D printed frame and positioned at a fixed location within the camera’s field of view.

A dedicated calibration mode is implemented in the GUI, which launches a separate calibration script prior to normal operation. In this mode, a static image of the grid is captured, and the user manually selects the four corner points of the grid in the image. These image-space coordinates correspond to known physical coordinates defined by the grid geometry. Based on the selected point correspondences, a 3×3 homography matrix $\mathbf{H}$ is computed such that

$$\mathbf{X} \sim \mathbf{H}\mathbf{x},$$

where $\mathbf{x}$ denotes homogeneous image coordinates and $\mathbf{X}$ denotes coordinates in the rectified substrate plane. The homography is calculated using standard Direct Linear Transformation (DLT) methods and is stored for subsequent use during runtime [8].

Figure 7.7 shows the GUI used for the calibration procedure. The calibration is performed once before normal operation to establish a consistent and accurate mapping between image coordinates and physical dimensions during runtime. Before initiating calibration, the system performs a homing and leveling sequence using the BLTouch sensor to ensure a repeatable reference plane. Once leveling is complete, the gantry automatically moves to a predefined position above the center of the bed at a fixed height and distance relative to the grid template. This positioning ensures that the calibration grid lies fully within the camera's field of view under consistent imaging conditions. After the camera stream is activated, the calibration process is initiated through the GUI. The user is prompted to manually select four corner points on the reference grid in a predefined order. The physical dimensions corresponding to these points are specified within the GUI based on the known grid spacing of 1 mm. As a result, selecting points separated by a known number of grid intervals directly defines the real-world distances used for homography computation. For example, a separation of 30 mm corresponds to 30 grid spacings. While this method provides flexibility, it requires manual counting of grid intervals during point selection. To improve usability and reduce the potential for human error, a future refinement is proposed in which a calibration template with pre-marked reference points is used. Calibration would then be performed using this dedicated template, which could subsequently be exchanged for an unmarked grid during normal operation. This approach would simplify the calibration procedure while preserving measurement accuracy.

The calibration controls, point selection order, and visual feedback provided by the GUI are illustrated in Figure 7.7.

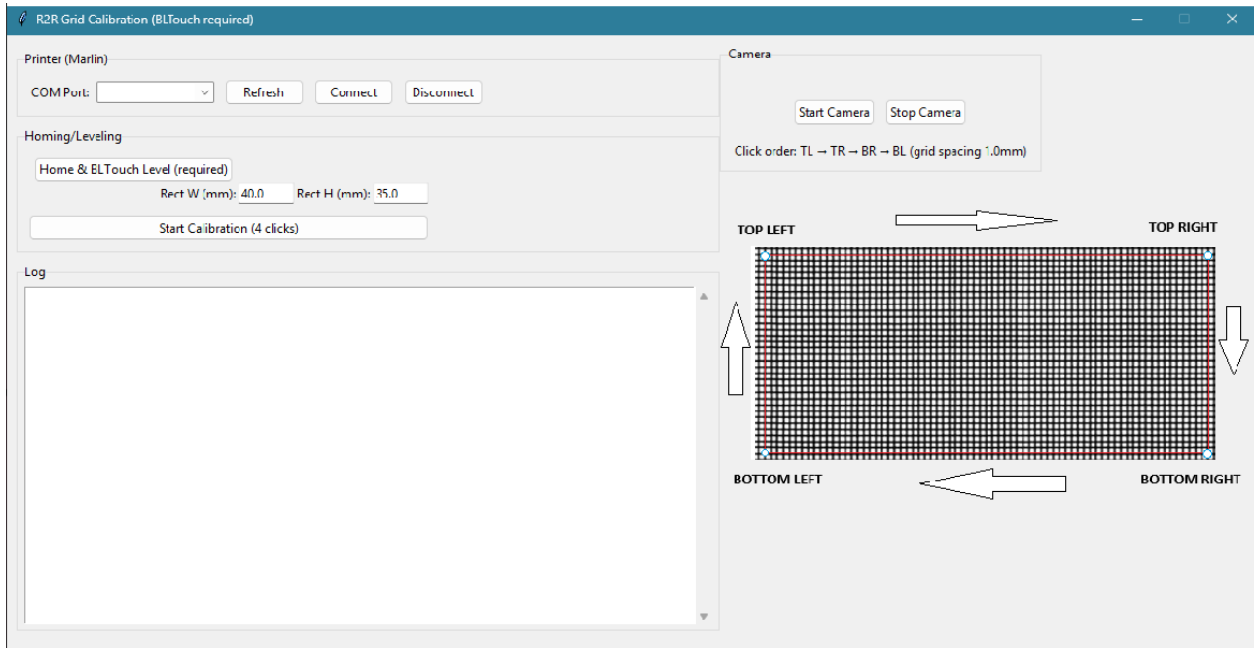


Figure 7.7: Calibration GUI

During normal operation, the computed homography is applied to each incoming frame to generate a rectified image in which distances are expressed in consistent physical units. Droplet detection and size estimation are then performed in this transformed space, allowing geometric measurements to remain invariant to camera angle and perspective effects. By separating calibration from runtime inference and maintaining a fixed spatial relationship between the grid, camera, and substrate, the system achieves repeatable and accurate droplet measurements within the R2R process.

7.6.2 Running AI on the R2R system

After completing calibration, the AI-based droplet detection system was deployed and evaluated on the full R2R setup. Figure 7.9 shows the GUI during runtime operation, where live camera input, droplet detections, and measured droplet dimensions are visualized in real time.

The YOLO-based segmentation model used for detection was trained exclusively on data collected from the prototype test setup. At the time of integration, the droplet dispensing module on the full R2R system was not operating reliably, which prevented the collection of a representative training dataset directly from the final machine configuration. As a result, the pretrained model was applied without additional retraining or fine-tuning on R2R-specific data. Despite being trained in a different experimental environment, the model was able to successfully detect and localize droplets on the R2R system. Combined with the homography-based calibration, the system was capable of estimating droplet dimensions in physical units during runtime. This demonstrates a degree of robustness to domain shift, as the model generalizes across changes in illumination, background appearance, and mechanical configuration between the prototype and R2R setups. Figure 7.8 shows the manually made droplet using a pipette to replicate the output of the dispenser if the implementation was successful.

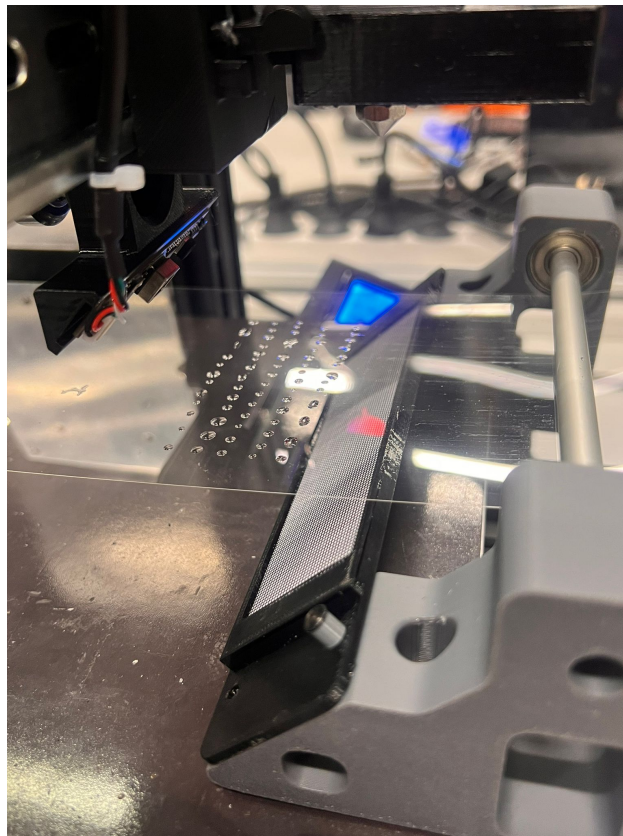


Figure 7.8: Testing AI droplet detection on R2R

For real-time operation, the camera stream and YOLO inference were executed on a laptop equipped with a dedicated GPU. Under this configuration, the system achieved a stable processing rate of approximately 30 FPS, enabling continuous and responsive droplet detection during scanning. In contrast, when the same pipeline was executed using CPU-only inference, the processing speed was limited to approximately 5–7 FPS on average, which was insufficient for reliable real-time monitoring.

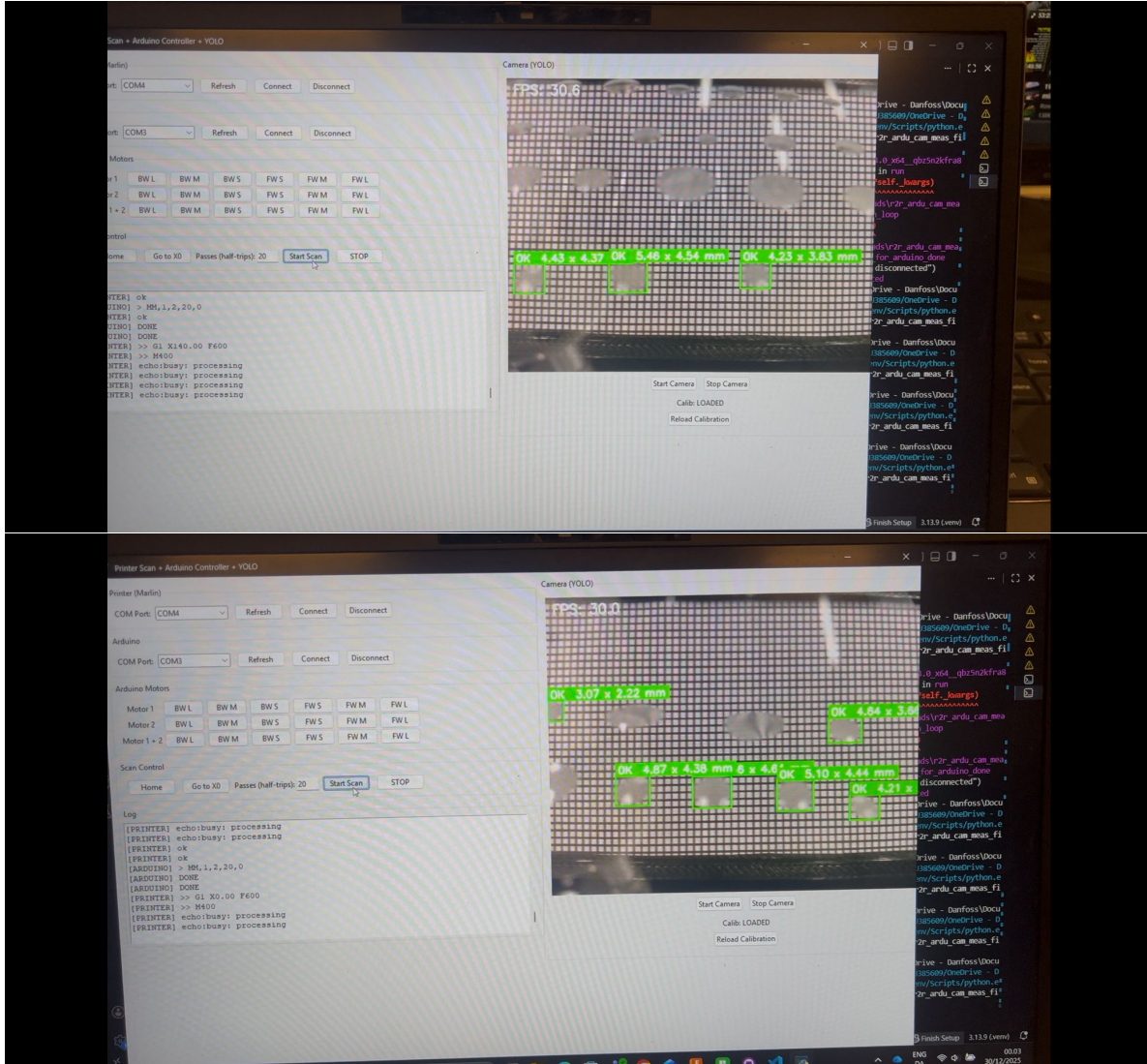


Figure 7.9: Output from GUI while running AI droplet detection

As a result, the availability of a dedicated GPU represents a practical requirement for successful real-time deployment of the proposed detection system. Without hardware acceleration, inference latency increases significantly, reducing resolution and increasing the likelihood of missed or unstable detections. This limitation highlights the importance of computational resources when integrating deep learning-based vision systems into R2R processes.

It is expected that collecting training data directly from the fully integrated R2R setup would further improve detection performance. Training or fine-tuning the model on environment-specific data would likely result in faster inference convergence, increased measurement consistency, and improved robustness against noise and variations in droplet morphology. Nonetheless, the current results demonstrate that the proposed AI-based approach remains functional and effective even under suboptimal training conditions.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

Despite continued efforts, a fully functional droplet dispensing module was not completed within the project timeframe. This limitation affected the progress of several interconnected parts of the system and restricted full end-to-end validation of the R2R Spotter Plotter. Nevertheless, the mechanical architecture, attachment methods, and overall system layout were designed from the beginning with modularity in mind, allowing for the integration of multiple dispensing modules in future iterations.

With respect to the computer vision component, the project demonstrated that AI-based droplet detection provides a fast and effective method for identifying and measuring droplets in real time. Even when trained on data collected from a prototype setup, the model was able to generalize to the full system configuration. It is expected that training and fine-tuning the model using data acquired directly from the final system would significantly improve detection accuracy, robustness, and consistency.

When considering the cost of the developed system, the project can be regarded as approaching its intended objectives. However, due to the incomplete implementation of the dispensing module, it is not possible to conclusively assess whether all performance and cost-related goals were fully satisfied.

In conclusion, the pneumatic approach to droplet dispensing within a R2R Spotter Plotter framework remains a viable concept. Further work would be required to re-evaluate the dispensing principle, refine the mechanical and fluidic design, and validate system performance. With these improvements, the proposed approach has the potential to serve as an effective alternative to existing dispensing solutions.

8.2 Future Work

8.2.1 DIN Rail Mount

Regarding the DIN Rail mount, during final testing the fragility and unreliability of the 3D printed pins was laid bare. When the pins are pulled, twisted or if the mount happens to fall on the ground, there is a risk that the pin itself could snap or deform, which in turn means that the entire DIN Rail mount needs to be reprinted, as the pin and spring is encased in the mount. Furthermore, the pins are a bit too loose in the mount, further risking damage to the pins when used. To substantially increase robustness, the pins could be made out of metal or any material that is stiffer than PLA, and be redesigned so that they fit more tightly in the DIN Rail mount and are kept straight when pulled.

The Gantry attachment vibrates too much at higher frequencies, increasing the risk of inconsistent droplet dispensing due to resonant frequencies. To combat this, tighter tolerances for the interacting parts can be used for the gantry attachment and DIN rail mount. Incorporating damping material such as rubber would also reduce frequencies.

8.2.2 Droplet Dispenser

The droplet dispenser was not fully functional, as it expelled liquid in a continuous spray rather than forming discrete droplets. To address this issue, future improvements should focus on selecting a spring with an appropriate spring constant and enabling independent control of the compression range without altering the spring's mechanical properties.

For improved droplet formation, the compression range must be significantly reduced and should be close to or less than 1 mm. In the current design, reducing the compression range also affects the effective spring constant, which limits precise control of the dispensing mechanism. Therefore, future designs should decouple these two parameters to allow independent adjustment of the spring constant and compression distance.

The calculation of a suitable spring for future improvements is presented in Appendix A.1. These calculations assume a coefficient of static friction between rubber and hard plastic of $\mu_s = 0.6$ [9, 10]. In addition, equations from the ORD 5700: O-Ring Handbook were used to determine the force that the spring must overcome. Only the larger O-ring was considered in the calculations, as the smaller O-ring experiences expansion only and does not contribute significantly to the resisting force. Based on these assumptions, the minimum required spring constant for a 1 mm deflection was calculated to be 24.4 N/mm. This value provides sufficient force to overcome static friction while preventing excessive compression, thereby addressing the issue of liquid spraying.

An alternative approach that could be explored is reversing the current actuation mechanism, using air pressure to push the needle downward and a spring to return it upward. This configuration may allow more precise control over the downward force applied to the needle, and consequently the force exerted on the liquid. Improved force control could reduce or eliminate the spraying behaviour observed in the current design.

8.3 AI Droplet Detection

Since the pneumatic dispensing unit was not operational by the end of the project, no droplet data from the final system configuration could be collected for training. As a result, the AI model was trained exclusively on data obtained from a test setup using manually generated droplets. While the AI still successfully detected most droplets, given this discrepancy, the AI was not trained for the fully correct conditions, which negatively affected detection accuracy.

An additional difference between the test setup and the final system is that, during operation, the camera moves while detection is performed. At sufficiently low speeds, this has a negligible impact; however, at higher speeds, motion blur is introduced. This degrades image quality and can distort the detected grid, further reducing model performance.

These limitations could be mitigated by including training data acquired under the same conditions, allowing the model to learn and compensate for these effects.

8.3.1 Electronic

There are two main issues with the current electrical design. First, the LM317 voltage regulator is used to step down the supply voltage from 24V to 12V, resulting in a voltage drop of 12V across the regulator. Since the LM317 supplies power to the Arduino, which draws approximately 0.18A, a significant amount of power is dissipated as heat within the regulator. As a result, thermal buildup occurs during prolonged operation, leading to increased temperature of the LM317 and potential reliability concerns.

$$P_{LM317} = (24V - 12V) \times I$$
$$\Delta T = P_{LM317} \times 50^\circ\text{C} = 108^\circ\text{C}$$

To address this issue, a switching buck converter such as the XL4015 can be used to replace the linear regulation stage. The buck converter would step the input voltage directly from 24V to 5V while supporting currents of up to 5A, significantly reducing power dissipation and thermal stress compared to the LM317-based solution. Secondly, when power is initially applied to the system, the motor draws a transient current

spike that can reach approximately 1A, as shown in Appendix C.1. This flow of current occurs during motor startup and can place additional stress on the power supply and switching components. The current spike can be mitigated using a simple soft-start circuit consisting of a series resistor and a MOSFET bypass. During power-up, the MOSFET remains off, forcing the current to flow through the resistor and limiting the inrush current. After a short delay, the MOSFET is switched on, bypassing the resistor and allowing normal operation.

Bibliography

- [1] Felix Leo Hoch. *R2R Spotter Plotter*. Bachelor Project. 2025.
- [2] Festo. *Solenoid Valves VUVS / Valve Manifold VTUS: Technical Data*. <https://docs.rs-online.com/5010/0900766b814f9ddb.pdf>. 2016.
- [3] Analog Devices. *TMC2225 Datasheet*. www.analog.com. 2020.
- [4] PBC linear. *NEMA17 Datasheet*. <https://media.pbclinear.com/>.
- [5] Eugene Hecht. *Optics*. 5th ed. Boston: Pearson Education, 2017.
- [6] Carsten Steger, Markus Ulrich, and Christian Wiedemann. *Machine Vision Algorithms and Applications*. Weinheim: Wiley-VCH, 2018.
- [7] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, 2004.
- [8] Gary Bradski. *OpenCV: Camera Calibration and 3D Reconstruction*. <https://docs.opencv.org>. 2023.
- [9] The Engineering ToolBox. *Friction Coefficients*. 2024. URL: https://www.engineeringtoolbox.com/friction-coefficients-d_778.html.
- [10] Bharat Bhushan. *Introduction to Tribology*. 2nd ed. Hoboken, NJ: John Wiley & Sons, 2013. ISBN: 978-1-118-40512-1.
- [11] Parker Hannifin Corporation. *Parker O-Ring Handbook*. ORD 5700. Accessed: 2 January 2026. Parker Hannifin Corporation. Cleveland, OH, USA, 2019. URL: <https://www.parker.com/Literature/O-Ring%20Division%20Literature/ORD%205700.pdf>.

Appendix A

Mechanical

A.1 Calculating the spring constant

The friction force generated by the large O-ring seal was estimated using a contact-pressure approach based on seal compression [11]. The O-ring cross-section diameter is:

$$d_{cs} = 2.4 \text{ mm}$$

The bore diameter at the seal location is:

$$D_b = 21.6 \text{ mm}$$

The radial groove depth is:

$$t = 2.15 \text{ mm}$$

The radial squeeze of the O-ring is given by:

$$\delta = d_{cs} - t$$

$$\delta = 2.4 - 2.15 = 0.25 \text{ mm}$$

The corresponding squeeze ratio is:

$$\varepsilon = \frac{\delta}{d_{cs}}$$

$$\varepsilon = \frac{0.25}{2.4} = 0.104 \approx 10.4\%$$

For an NBR O-ring with a hardness of 70 Shore A, the Young's modulus is approximated as [11]:

$$E \approx 8 \text{ MPa}$$

The contact pressure is estimated using a linear elastic approximation:

$$p_c \approx E\varepsilon$$

$$p_c \approx 8 \times 0.104 = 0.83 \text{ MPa}$$

Contact Area

The contact width is estimated as a fraction of the O-ring cross-section:

$$w \approx 0.3d_{cs}$$

$$w \approx 0.3 \times 2.4 = 0.72 \text{ mm}$$

The contact area between the O-ring and the bore is:

$$A_c = \pi D_b w$$

$$A_c = \pi(21.6)(0.72) = 48.9 \text{ mm}^2 = 4.89 \times 10^{-5} \text{ m}^2$$

The normal force acting on the O-ring is:

$$N = p_c A_c$$

$$N = (0.83 \times 10^6)(4.89 \times 10^{-5}) = 40.6 \text{ N}$$

The assumed static coefficient of friction:

$$\mu_s \approx 0.6$$

The resulting static friction force is:

$$F_s = \mu_s N$$

$$F_s = 0.6 \times 40.6 = 24.4 \text{ N}$$

The estimated breakaway friction force generated by the large O-ring is therefore:

$$F_{s,\text{big}} \approx 24 \text{ N}$$

The spring force is governed by Hooke's law:

$$F = kx$$

For a compression of:

$$x = 1 \text{ mm}$$

Using the estimated static friction force from the large O-ring:

$$F_{s,\text{big}} = 24.4 \text{ N}$$

the minimum spring constant required to overcome this friction is:

$$k = \frac{F_{s,\text{big}}}{x}$$

$$k = \frac{24.4}{1} = 24.4 \text{ N/mm}$$

The ideal spring constant required to overcome the estimated static friction force with a compression of 1 mm is therefore:

$$\boxed{k_{\text{ideal}} \approx 24 \text{ N/mm}}$$

A.2 Droplet dispenser

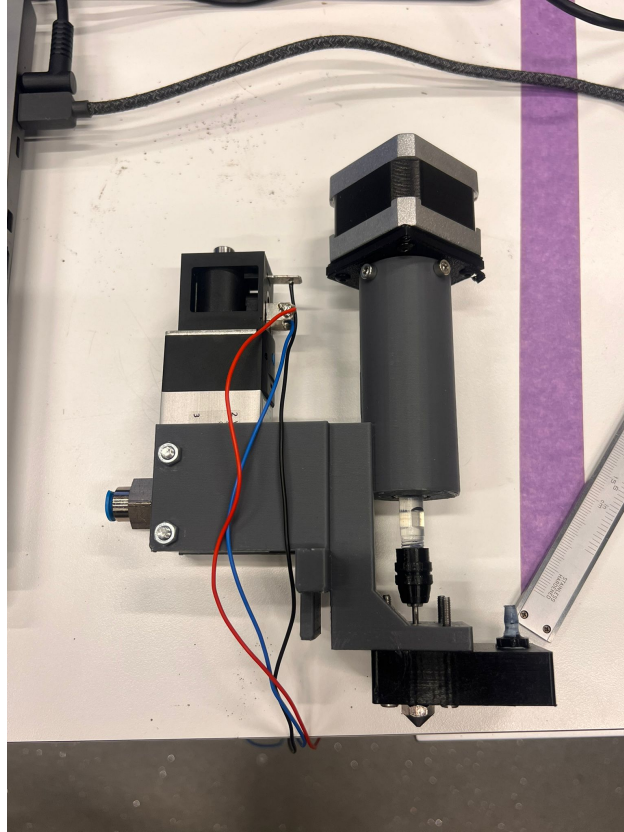


Figure A.1: Droplet dispenser

A.3 R2R Serial Plotter

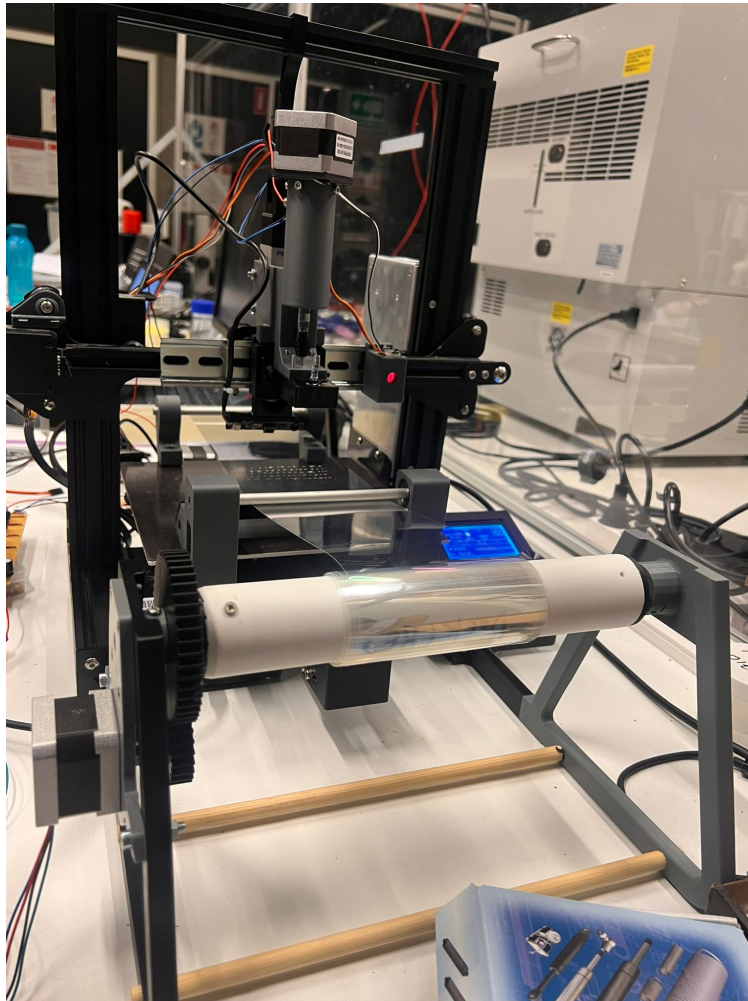


Figure A.2: Full R2R assembly

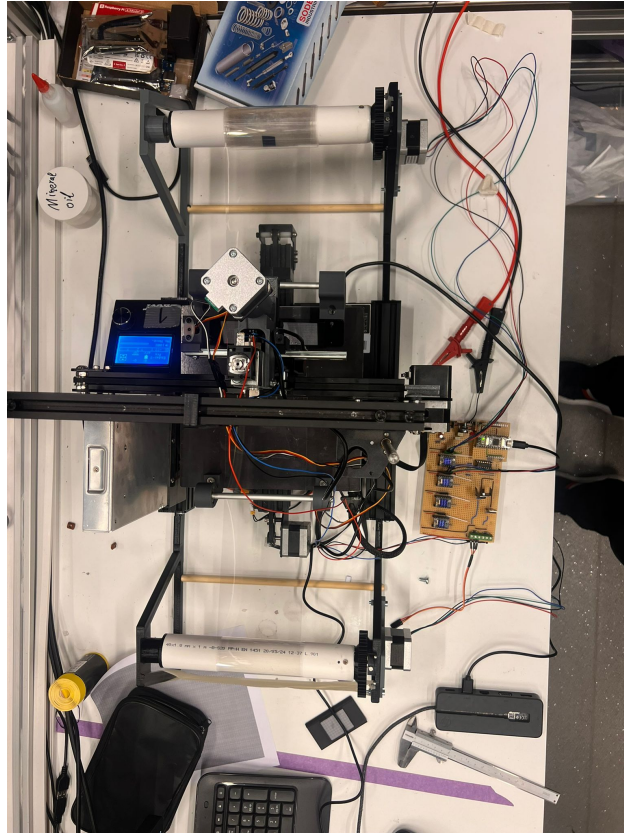


Figure A.3: Top view of whole assembly during testing

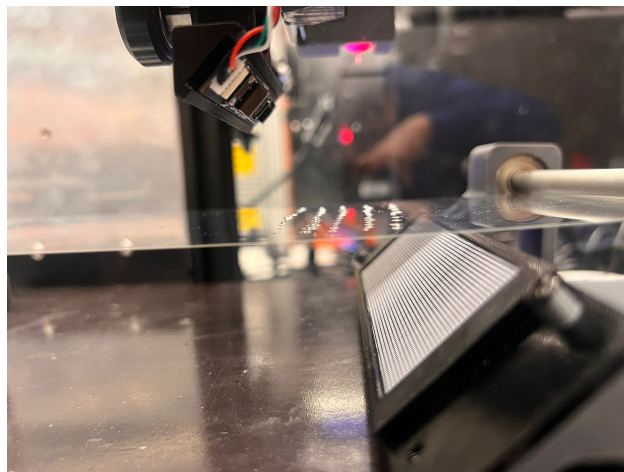


Figure A.4: Camera and grid on R2R

Appendix B

Software and Machine Learning

B.1 GUI output while testing the AI droplet detection

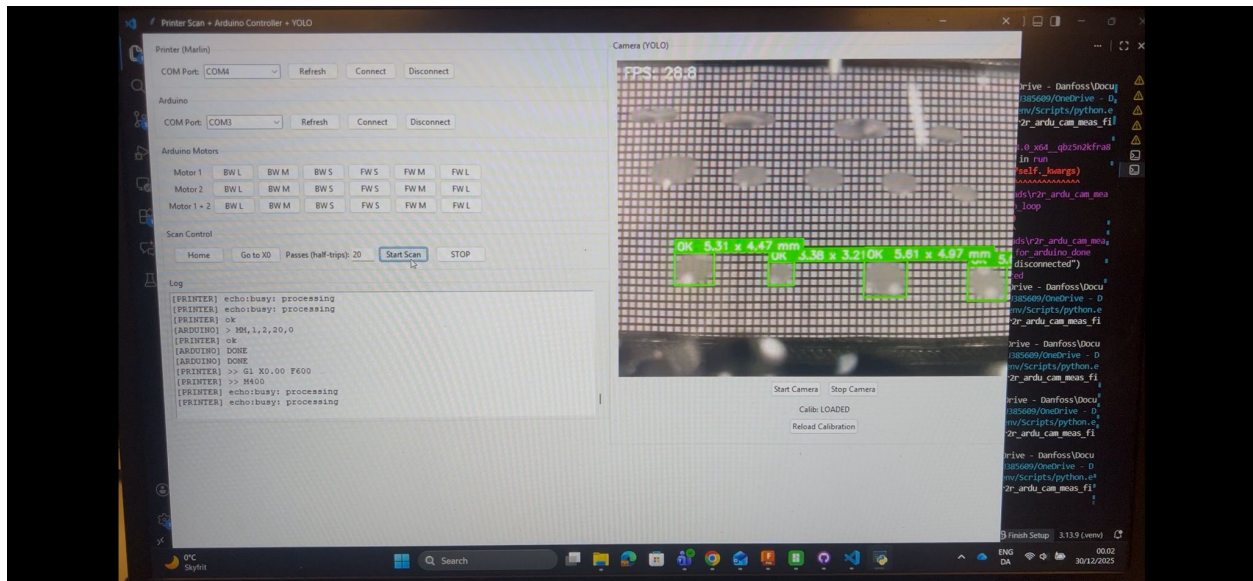


Figure B.1: AI Droplet testing

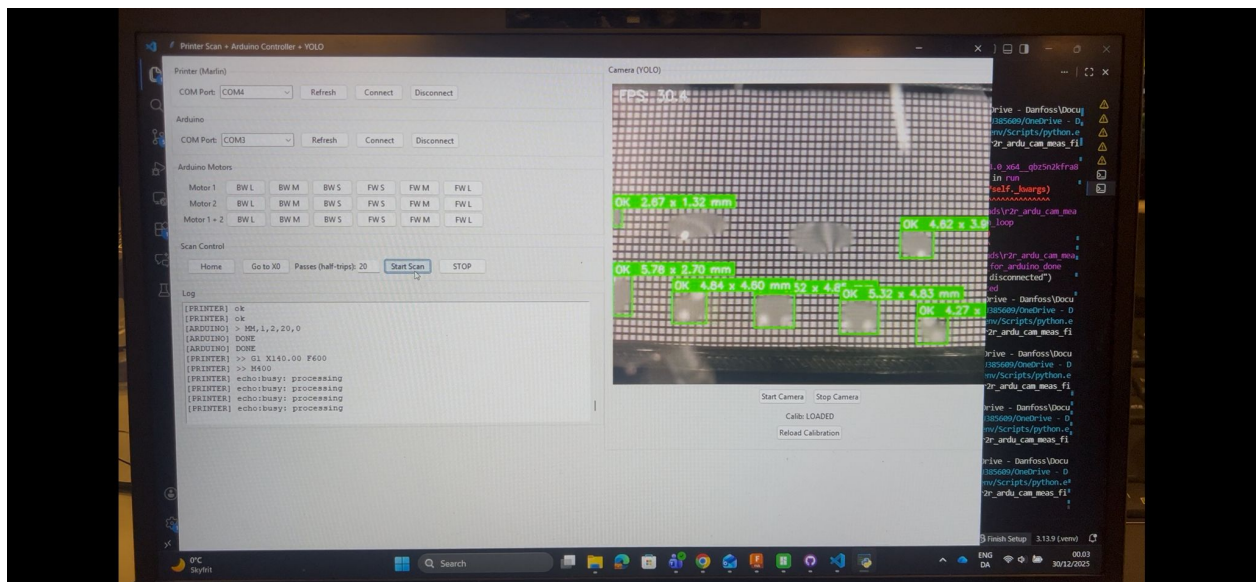


Figure B.2: AI Droplet testing

Appendix C

Electronics

C.1 Current shown in motor when powering up

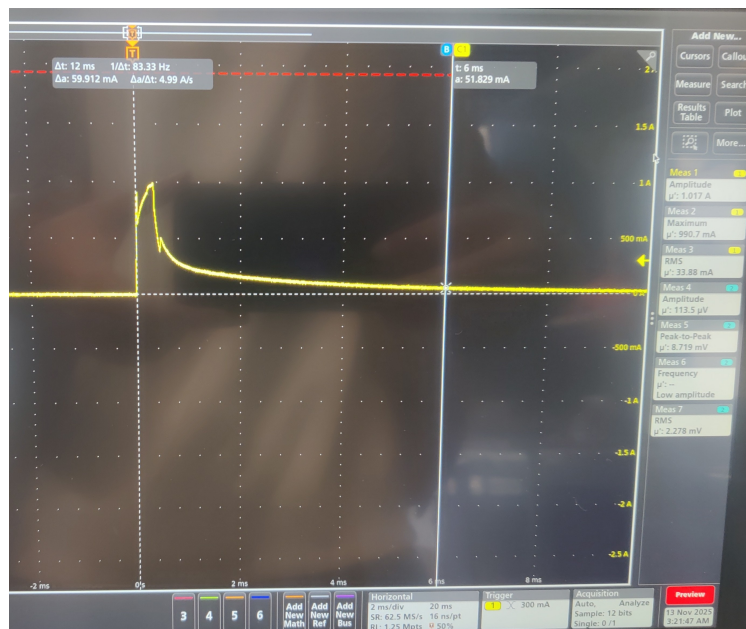


Figure C.1: Current amount when powering up motor